

Urban Monitoring Services for One Health

Sistemas Distribuídos - Practical Assignment nº1

Phase 1

Authors:

João Lopes --Al81202

Pedro Costa-- Al81940

Zhera --Al2025174403

Communication Protocol for Urban Monitoring Services

1. Introduction

The proposed system aims to simulate a distributed urban environmental monitoring infrastructure based on the One Health paradigm, which recognizes the strong interconnection between human, animal, and environmental health.

The system architecture is composed of three main entities:

SENSOR – responsible for collecting environmental data;

GATEWAY – responsible for receiving, validating, and forwarding data;

SERVER – responsible for storing and processing the collected information.

Communication between these entities is carried out through TCP sockets, using a text-based message protocol.

2. System Flow Simulation

2.1 Identification and Registration Phase

The sensor starts communication with the gateway through an identification message. After verifying that the sensor is registered, the gateway confirms the connection. Then, the sensor sends the list of data types it is able to collect, which is validated by the gateway.

SENSOR -> GATEWAY: HELLO | S102

GATEWAY -> SENSOR: OK

SENSOR -> GATEWAY: TYPES | TEMP, PM2.5

GATEWAY -> SENSOR: OK

2.2 Environmental Data Transmission Phase

After the registration process is completed, the sensor can send collected environmental measurements. The gateway receives the message, validates the sensor and the data type, and responds with an acknowledgment.

SENSOR -> GATEWAY: DATA | S102 | PM2.5 | 78 | 2026-03-10T09:15:00

GATEWAY -> SENSOR: ACK

2.3 Forwarding Phase to the Server

After validating the received measurement, the gateway forwards the information to the server, adding the geographical zone associated with the sensor. The server stores the data and sends a confirmation back to the gateway.

GATEWAY -> SERVER: STORE | S102 | ZONA_ESCOLAR | PM2.5 | 78 | 2026-03-10T09:15:00
SERVER -> GATEWAY: STORED

2.4 Video Request Phase

During operation, the sensor may request the creation of a video stream. The gateway receives the request, validates whether the sensor supports video transmission, and sends a confirmation message.

SENSOR -> GATEWAY: VIDEO_REQUEST | S102
GATEWAY -> SENSOR: ACK

2.5 Heartbeat Phase

During operation, the sensor periodically sends heartbeat messages to indicate that it is still active. The gateway responds by confirming that the sensor remains connected.

SENSOR -> GATEWAY: HEARTBEAT | S102
GATEWAY -> SENSOR: ALIVE

2.6 Termination Phase

When the sensor intends to end the communication, it sends a disconnection message to the gateway. The gateway confirms the termination of the connection.

SENSOR -> GATEWAY: DISCONNECT | S102
GATEWAY -> SENSOR: BYE